

Literal Low-Window Affine Export: Content-Factor Extraction, Resonance Sectors, and Fixed-Determinant Preservation

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Abstract

Let A_X be the already aggregated fixed- h_0 determinant coefficient of TPC-32, embedded without wrap in $\mathbb{Z}/q_X\mathbb{Z}$, and let $\mathbf{m}_{K,X}$ be the order- K zero-arc reconstruction multiplier of TPC-86. We export the complete low-frequency functional

$$\mathcal{L}_{K,R}(A_X) = \sum_{0 < d_{q_X}(r,0) \leq R} \mathbf{m}_{K,X}(r) \hat{A}_X(r)$$

to the native affine variables of the two polarized physical columns. The first exact formula is row-native:

$$\mathcal{L}_{K,R}(A_X) = \sum_{\alpha \neq \gamma} \sum_j w_{\alpha,\gamma,j} \mathbf{1}_{G_{\alpha,\gamma}(j) \leq C_X} \mathcal{K}_{K,R,X} \left(\frac{m_\alpha - m_\gamma}{G_{\alpha,\gamma}(j)} \right),$$

where $\mathcal{K}_{K,R,X}(y) = \sum_{0 < d_q(r,0) \leq R} \mathbf{m}_{K,X}(r) e_q(-ry)$. No frequency, channel, or exceptional sector has been estimated or discarded.

After opening one ultra leg and applying the actual row-mask projector, every left column has

$$D_\theta(t) = d_\theta + \sigma_\theta t, \quad U_\theta(t) = u_\theta + a_\theta t, \quad \sigma_\theta u_\theta - a_\theta d_\theta = h_0,$$

with the exact two-Möbius sign $\mu(D_\theta(t))\mu(U_\theta(t))$; the right polarization is symmetric. We prove a decorated-column principle showing that the small-content indicator and determinant character survive this pointwise disintegration. Exact content inversion then gives the complete multiplier-weighted expansion

$$\sum_{\theta,c,\kappa,r} c_{\theta,X} \mu(\kappa) \mathbf{m}_{K,X}(r) \mathbf{1}_{c\kappa|P_\theta} \sum_{t \in I_\theta} \mu(D_\theta(t)) \mu(U_\theta(t)) W_\theta(t) \mathbf{1}_{c\kappa|\sigma_\theta U_\theta(t)} e_{cq_X}(-r\varepsilon_\theta(M_\theta(t) - n_\theta)).$$

All masks and provenance factors are retained inside $c_{\theta,X}$ and W_θ . An explicit child-to-source inverse and projector-weighted multiplicity identity prove that this key list reassembles the literal source coefficient.

Writing $b = c\kappa$,

$$B_{\theta,b} = \frac{b}{(b, \sigma_\theta)},$$

the divisibility condition either has no solution or restricts t to one class $t = \tau + B_{\theta,b}z$. On that class the determinant character has slope $\ell_\theta v_\theta \sigma_\theta B_{\theta,b}$. The unreduced forms

$$D_\theta(\tau + Bz), \quad U_\theta(\tau + Bz)$$

have determinant Bh_0 , but the divisibility condition also gives

$$U_\theta(\tau + Bz) = B V_{\theta,b}(z).$$

On the nonzero Möbius support, B is squarefree and

$$\mu(BV) = \mu(B)\mu(V)\mathbf{1}_{(B,V)=1}.$$

After this exact content-factor extraction, the reduced pair

$$D_{\theta,b}(z), \quad V_{\theta,b}(z)$$

has slopes $(\sigma_\theta B, a_\theta)$ and determinant exactly

$$\boxed{h_0}.$$

Physical primitivity also gives $(a_\theta, \sigma_\theta B) = 1$ and $(a_\theta \sigma_\theta B, h_0) = 1$. Thus the low window can be resolved into unmasked fixed- h_0 affine intervals, but only after retaining the extracted sign $\mu(B)$ and the new coprimality mask. Treating the unreduced Bh_0 pair as though its determinant were h_0 is the precise provenance error.

For every threshold H_X , we finally partition every resolved native term, without deletion, into short fibers, long resonant fibers satisfying

$$N_\xi \left\| \frac{r_\xi \ell_\xi v_\xi \sigma_\xi B_\xi}{c_\xi q_X} \right\|_{\mathbb{R}/\mathbb{Z}} \leq 1,$$

and the complementary long generic fibers. This supplies the complete representation requested by TPC-92 Deliverable 1A and identifies the exact arithmetic inputs still missing. The finite reindexing and determinant law are L0; their attachment to the literal packet is L1. No L2 cancellation, fixed-shift prime-pair lower bound, parity breach, or twin-prime conclusion is proved.

Keywords: low-frequency window; determinant character; affine Möbius column; exact content; resonance; short fiber; provenance; fixed shift.

Literal-frame convention. Every physical formula retains one prescribed $h_0 \neq 0$, the actual small-content coefficient, both cutoff polarizations, every native row, divisor, interval, mask and separation key, and one global post-aggregation normalization. No statement is specialized to $h_0 = 2$.

1 Introduction

TPC-86 reduces the distinguished zero-mode problem to a coherent low-frequency window for the literal normalized-determinant coefficient [3]. At the critical row,

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}, \quad N_{0,X} = J_X Q_X^2,$$

and an order- K zero-arc filter gives

$$|Z_X| \ll_K \left| \sum_{0 < d_{q_X}(r,0) \leq R_X} \mathbf{m}_{K,X}(r) \hat{A}_X(r) \right| + X^{o(1)} N_{0,X} R_X^{1/2-K}. \quad (1)$$

The minimal arithmetic input is the signed multiplier-weighted sum, not a maximum over generic frequencies. TPC-92 therefore lists a literal export of that sum as its first next-stage deliverable [4].

The word “export” carries more content than inserting an additive character in an abstract affine sum. The physical coefficient has:

- two polarized cutoff legs;
- an exact common-target content $G \leq C_X = \lfloor J_X \rfloor$;
- a normalized determinant $(m_\alpha - m_\gamma)/G$;
- a joint row-pair mask;
- an opposite row fixed outside each affine column;
- row-gcd divisor layers and fixed-period factors;
- long and bounded-length affine fibers; and

- one outer reassembly under the original normalization.

Every one of these data affects either the phase or the coefficient.

This paper performs that export exactly. It also identifies a content-factor extraction that is invisible at determinant frequency zero. Once exact content inversion imposes

$$c\kappa \mid \sigma U(t),$$

solving the divisibility condition restricts the affine parameter to one progression $t = \tau + Bz$. The unreduced reparametrized pair has determinant Bh_0 , but the same divisibility also gives

$$U(\tau + Bz) = BV(z).$$

Extracting this literal content factor restores determinant h_0 :

$$\begin{aligned} (D(\tau + Bz), U(\tau + Bz)) &: \det = Bh_0, \\ (D(\tau + Bz), V(z)) &: \det = h_0. \end{aligned}$$

On the nonzero Möbius support, the extraction contributes the literal factor $\mu(B)$ and the mask $\mathbf{1}_{(B,V)=1}$. Dropping either one is an invalid hybrid; retaining both gives a genuine fixed- h_0 affine family.

1.1 Main results

The exact chain has four steps.

Row-native window. We first keep the physical rows and prove that the complete low-window functional is obtained by replacing the determinant character with one finite kernel

$$\mathcal{K}_{K,R,X}(y) = \sum_{0 < d_{qX}(r,0) \leq R} \mathfrak{m}_{K,X}(r) e_{qX}(-ry).$$

This identity already returns every low frequency with the correct multiplier and phase.

Decorated affine columns. The TPC-33 reflection and row-gcd projection are pointwise identities [2]. We prove that they remain valid after multiplication by the small-content determinant decoration. This produces an exact two-sign affine formula for both polarizations, with the opposite row and every separation key retained in the outer provenance tuple. The source-to-child construction, its inverse, and the projector-weighted child multiplicity are all displayed explicitly; losslessness is therefore a conclusion rather than an interface assumption.

Content-progression reduction. We expand exact content by the TPC-32 common-divisor projector [1]. The resulting congruence is solved explicitly and the two-stage determinant law

$$Bh_0 \longrightarrow h_0$$

is proved after exact extraction of the forced factor B from one affine value. This is an algebraic theorem, not a heuristic normalization.

Complete sector partition. Every resolved term is assigned to exactly one of three sets: short, long resonant, or long generic. Resonance is measured by the actual phase increment on the resolved progression. The partition is an equality. In particular, neither exact resonances nor near-resonant long fibers are hidden inside an exceptional “generic” remainder.

1.2 Proof levels

We use the established labels:

- L0 finite Fourier identities, exact projector expansions, affine reparametrization, determinant laws, and set partitions;
- L1 an exact attachment to the same fixed- h_0 TPC-32 coefficient, with its full native key list and global normalization;
- L2 a new growing- X cancellation estimate for the resulting literal window or one of its returned sectors.

The new theorems are L0, and the export is L1. No L2 estimate is asserted.

1.3 Organization

Section 2 gives the exact row-native window. Section 3 proves the decorated-column principle and the native affine formula. Section 4 resolves exact content. Section 5 proves fixed-determinant preservation and the provenance theorem. The short/resonant/generic partition appears in section 6. Section 7 records the remaining arithmetic estimates and endpoint ledgers, and section 8 states the audited conclusion.

2 The complete row-native low window

2.1 The literal coefficient

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$. A physical row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q_X,$$

and

$$N_\alpha(j) = m_\alpha j + h_0.$$

For a second row γ , put

$$G_{\alpha,\gamma}(j) = \gcd(N_\alpha(j), N_\gamma(j)), \quad \Delta_{\alpha,\gamma}^\#(j) = \frac{m_\alpha - m_\gamma}{G_{\alpha,\gamma}(j)}. \quad (2)$$

Primitive target support makes the last quantity a nonzero integer on the off-diagonal physical support.

Let

$$A_{m,Y}(j) = \sum_{\substack{w \leq Y \\ w|mj+h_0}} b_R(w), \quad C_m(j) = A_{m,U_0}(j) - A_{m,T}(j),$$

and

$$H_m(j) = \frac{A_{m,U_0}(j) + A_{m,T}(j)}{2}.$$

The exact two-leg polarization is

$$\mathcal{K}_{\alpha,\gamma}^{\text{sh}}(j) = C_{m_\alpha}(j)H_{m_\gamma}(j) + H_{m_\alpha}(j)C_{m_\gamma}(j). \quad (3)$$

Define the complete signed atom

$$w_{\alpha,\gamma,j} = \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathfrak{A}_{\alpha,\gamma}(j) \mathcal{K}_{\alpha,\gamma}^{\text{sh}}(j). \quad (4)$$

The already aggregated determinant coefficient is

$$A_X(n) = \sum_{\substack{\alpha \neq \gamma \\ j}} w_{\alpha,\gamma,j} \mathbf{1}_{G_{\alpha,\gamma}(j) \leq C_X} \mathbf{1}_{\Delta_{\alpha,\gamma}^\#(j)=n}, \quad C_X = \lfloor J_X \rfloor. \quad (5)$$

The sum in (5) is taken before any norm or Fourier estimate.

Choose the strong no-wrap prime modulus $q_X \asymp Q_X$ from TPC-86 and identify A_X with its zero-padded cyclic coefficient. Then

$$\hat{A}_X(r) = \sum_{\substack{\alpha \neq \gamma \\ j}} w_{\alpha, \gamma, j} \mathbf{1}_{G_{\alpha, \gamma}(j) \leq C_X} e_{q_X} \left(-r \Delta_{\alpha, \gamma}^{\#}(j) \right). \quad (6)$$

2.2 The reconstruction multiplier

Fix an integer $K \geq 2$. Let $p_{K, X}$ be the chosen translated K -fold interval probability supported inside the positive-density zero arc of the padded coefficient, and define

$$\mathbf{m}_{K, X}(r) = \sum_{x \pmod{q_X}} p_{K, X}(x) e_{q_X}(rx). \quad (7)$$

TPC-86 proves

$$Z_X = - \sum_{r \neq 0} \mathbf{m}_{K, X}(r) \hat{A}_X(r), \quad |\mathbf{m}_{K, X}(r)| \ll_K (1 + d_{q_X}(r, 0))^{-K}.$$

For $1 \leq R < q_X/2$, put

$$\mathscr{W}_X(R) = \{r \in \mathbb{Z}/q_X\mathbb{Z} : 0 < d_{q_X}(r, 0) \leq R\}, \quad (8)$$

$$\mathcal{L}_{K, R}(A_X) = \sum_{r \in \mathscr{W}_X(R)} \mathbf{m}_{K, X}(r) \hat{A}_X(r), \quad (9)$$

$$\mathcal{K}_{K, R, X}(y) = \sum_{r \in \mathscr{W}_X(R)} \mathbf{m}_{K, X}(r) e_{q_X}(-ry). \quad (10)$$

Because $R < q_X/2$, every class in $\mathscr{W}_X(R)$ has a unique signed lift $\tilde{r} \in \mathbb{Z}$ with $0 < |\tilde{r}| = d_{q_X}(r, 0) \leq R$. Whenever an ordinary integer divisibility involving a frequency is written below, we use this lift and suppress the tilde.

Theorem 2.1 (Exact row-native low-window export). *For the literal coefficient (5),*

$$\boxed{\mathcal{L}_{K, R}(A_X) = \sum_{\substack{\alpha \neq \gamma \\ j}} w_{\alpha, \gamma, j} \mathbf{1}_{G_{\alpha, \gamma}(j) \leq C_X} \mathcal{K}_{K, R, X} \left(\Delta_{\alpha, \gamma}^{\#}(j) \right)}. \quad (11)$$

Moreover,

$$Z_X = -\mathcal{L}_{K, R}(A_X) + \mathcal{E}_{K, R, X}, \quad |\mathcal{E}_{K, R, X}| \ll_K X^{o(1)} N_{0, X} R^{1/2-K}. \quad (12)$$

Proof. Insert (6) into (9) and interchange the finite sums. The inner frequency sum is exactly (10), proving (11). Split the exact TPC-86 reconstruction at $d_{q_X}(r, 0) = R$. Its Cauchy–Schwarz tail estimate, together with the inherited upper bound $q_X D_X \ll X^{o(1)} N_{0, X}^2$, gives (12). $\square$

Remark 2.2 (Why this is already a complete window). [Theorem 2.1](#) sums the two polarizations through (3), uses the exact content rather than a majorant, and retains every frequency in $\mathscr{W}_X(R)$ with its original multiplier. It is not a maximum over r , a generic-frequency estimate, or an absolute sum over physical atoms.

Remark 2.3 (Filter choice). Different translations of the zero-arc filter change $\mathbf{m}_{K, X}(r)$ by a phase. The theorem holds for the one filter fixed in (7). Replacing it by its absolute value changes the coherent functional and is not an identity.

3 Decorated affine-column export

The TPC-33 column theorem was stated for the untwisted distinguished shell. Its proof uses pointwise row-gcd inversion, divisor reflection, and sign fusion. These operations commute with any scalar decoration attached to the original physical atom. We record that principle before applying it to the determinant character.

3.1 A pointwise decoration lemma

Lemma 3.1 (Decorated reindexing). *Let a finite sum be written as*

$$S = \sum_{\omega \in \Omega} a(\omega),$$

and suppose a finite exact expansion and reindexing gives

$$a(\omega) = \sum_{\xi \in \Xi(\omega)} a_\xi(\omega), \quad \{(\omega, \xi) : \omega \in \Omega, \xi \in \Xi(\omega)\} \longleftrightarrow \mathcal{P}.$$

For every function $F : \Omega \rightarrow \mathbb{C}$,

$$\sum_{\omega \in \Omega} F(\omega) a(\omega) = \sum_{p \in \mathcal{P}} F(\omega(p)) a_p.$$

In particular, a pointwise projector or change of variables remains exact after multiplication by F , provided the provenance map $p \mapsto \omega(p)$ is retained.

Proof. Multiply the finite identity for $a(\omega)$ by $F(\omega)$, sum over ω , and use the stated bijective reindexing. No estimate or interchange of an infinite sum is involved. $\square$

Remark 3.2 (Why provenance is part of the statement). The conclusion fails if two distinct source atoms are merged before the value of F is known. Equal affine forms or equal determinants do not imply equal determinant phases: the opposite row, exact content, and source polarization remain part of $\omega(p)$.

3.2 Lossless native keys

Apply the exact polarization (3). For the left-ultra leg, fix the opposite row

$$\gamma = (\ell', e), \quad n_\gamma = \ell' e,$$

open an ultra divisor u , and write

$$\ell d j + h_0 = u \sigma.$$

On a row-gcd layer $v \mid d, e$, write $d = vD$. The TPC-33 reflection gives

$$D_\theta(t) = d_\theta + \sigma_\theta t, \quad U_\theta(t) = u_\theta + a_\theta t, \quad a_\theta = \ell_\theta j_\theta v_\theta, \quad (13)$$

with

$$\sigma_\theta u_\theta - a_\theta d_\theta = h_0, \quad (a_\theta, \sigma_\theta) = 1, \quad (a_\theta \sigma_\theta, h_0) = 1. \quad (14)$$

The affine parameter ranges over one of at most two intervals I_θ , and

$$|I_\theta| \ll 1 + \frac{D_X^{\text{row}}}{\sigma_\theta v_\theta}, \quad (15)$$

where D_X^{row} denotes the dyadic row scale, not the determinant energy.

To avoid suppressing data, define the left provenance key

$$\theta = (L, \gamma, \ell, j, \sigma, v, \iota). \quad (16)$$

Here L records the polarization, ι records which of the at most two interval components occurs, and v is the actual row-gcd-projector divisor. The fixed-period factor and the literal smooth, logarithmic, Mellin/BV, residue, and source-prefix factors are evaluated as functions in ρ_θ and $W_{\theta, X}$; they are not replaced by a new anonymous key. Right-ultra keys have the symmetric form with α fixed and are marked by R .

For a left key, set

$$n_\theta = n_\gamma = \ell' e, \quad (17)$$

$$M_\theta(t) = \ell_\theta v_\theta D_\theta(t), \quad (18)$$

$$P_\theta = n_\theta j_\theta + h_0, \quad (19)$$

$$G_\theta(t) = \gcd(\sigma_\theta U_\theta(t), P_\theta). \quad (20)$$

Define the determinant orientation by

$$\varepsilon_\theta = \begin{cases} +1, & \theta \text{ is left-ultra,} \\ -1, & \theta \text{ is right-ultra.} \end{cases} \quad (21)$$

Thus the ordered row determinant inherited from (α, γ) is always

$$\Delta_\theta^\#(t) = \varepsilon_\theta \frac{M_\theta(t) - n_\theta}{G_\theta(t)}. \quad (22)$$

The equality

$$\sigma_\theta U_\theta(t) = M_\theta(t) j_\theta + h_0$$

holds identically. Define the retained local mask

$$\chi_\theta(t) = \mathbf{1}_{(v_\theta, D_\theta(t) U_\theta(t))=1} \rho_\theta(t), \quad (23)$$

where ρ_θ is the fixed-period factor from the actual-mask column theorem. Let $W_{\theta,X}(t)$ contain the remaining literal smooth and logarithmic factors, and let $c_{\theta,X}$ contain the opposite-row coefficient and all factors constant on the affine fiber. These are the literal TPC-33 factors evaluated on the source atom reconstructed below; no new fiber normalization is introduced.

3.3 The source–child inverse and projector multiplicity

Let $G_X^{\text{row}} = X^{\kappa_0}$ denote the row-gcd cutoff of the actual TPC-33 mask and put

$$\lambda_{G_X^{\text{row}}}(v) = \sum_{\substack{g|v \\ g \leq G_X^{\text{row}}}} \mu(v/g).$$

A left preprojector source atom is

$$\omega = (L, \alpha, \gamma, j, u), \quad \alpha = (\ell, d), \quad \gamma = (\ell', e), \quad (24)$$

where u is one opened divisor in the left ultra increment and

$$\ell d j + h_0 = u \sigma.$$

For every algebraic child divisor $v \mid d, e$, choose the canonical representative $d_{\omega,v} \in \{0, \dots, \sigma - 1\}$ satisfying

$$\ell j v d_{\omega,v} + h_0 \equiv 0 \pmod{\sigma}, \quad (25)$$

and define

$$t_{\omega,v} = \frac{d/v - d_{\omega,v}}{\sigma}, \quad u_{\omega,v} = \frac{\ell j v d_{\omega,v} + h_0}{\sigma}. \quad (26)$$

Target primitivity gives $(\ell j v, \sigma) = 1$, so the residue is unique and $t_{\omega,v} \in \mathbb{Z}$. Use disjoint integer interval components for the retained near-diagonal set and let ι be the unique component containing $t_{\omega,v}$.

The corresponding child key has

$$d_\theta = d_{\omega,v}, \quad u_\theta = u_{\omega,v}, \quad \sigma_\theta = \sigma, \quad v_\theta = v.$$

For a right preprojector atom

$$\omega = (R, \alpha, \gamma, j, u), \quad \alpha = (\ell, d), \quad \gamma = (\ell', e),$$

the same construction is made after exchanging (α, ℓ, d) with (γ, ℓ', e) .

Theorem 3.3 (Explicit source–child reconstruction). *The preceding construction has the following properties.*

(i) *For every left child $p = (\theta, t)$, the inverse provenance map is*

$$\omega(p) = (L, (\ell_\theta, v_\theta D_\theta(t)), \gamma_\theta, j_\theta, U_\theta(t)). \quad (27)$$

For every right child it is

$$\omega(p) = (R, \alpha_\theta, (\ell_\theta, v_\theta D_\theta(t)), j_\theta, U_\theta(t)). \quad (28)$$

(ii) *For a fixed source atom and fixed $v \mid d, e$, $(\theta, t_{\omega, v})$ is its unique child on that projector layer. Hence the algebraic source multiplicity is exactly $\tau((d, e))$; the interval component introduces no further multiplicity.*

(iii) *The multiplicity is projector-weighted, not a duplication of the physical atom:*

$$\sum_{v \mid d, e} \lambda_{G_X^{\text{row}}}(v) = \mathbf{1}_{(d, e) \leq G_X^{\text{row}}}. \quad (29)$$

On the physical squarefree support the moving sign is reconstructed term by term as

$$\begin{aligned} & \mu(d)(-\mu(u) \log u) \mathbf{1}_{(d, e) \leq G_X^{\text{row}}} \\ &= \sum_{v \mid d, e} [-\mu(v) \lambda_{G_X^{\text{row}}}(v)] \mu(d/v) \mu(u) \log u \mathbf{1}_{(v, (d/v)u)=1}. \end{aligned} \quad (30)$$

(iv) *The labels L and R form a disjoint union and correspond to the two summands of (3). Thus both polarizations occur once before their intentional projector multiplicity.*

Consequently there is a concrete bijection

$$\{(\omega, v) : \omega \text{ is a retained source atom, } v \mid d_\omega, e_\omega\} \longleftrightarrow \{(\theta, t) : \theta \in \Theta_X, t \in I_\theta\}.$$

The forward map is (26), with its polarization and interval label, while (27)–(28) give its actual child-to-source inverse.

Proof. For a left source, $d = v(d/v)$ and

$$\ell j v(d/v) + h_0 = u\sigma.$$

Reduction modulo σ gives (25), so $d/v = d_{\omega, v} + \sigma t_{\omega, v}$. Therefore

$$v D_\theta(t_{\omega, v}) = d.$$

Using the affine determinant identity at the same parameter gives

$$\sigma U_\theta(t_{\omega, v}) = \ell j v D_\theta(t_{\omega, v}) + h_0 = u\sigma,$$

and hence $U_\theta(t_{\omega, v}) = u$. This proves (27); it also proves uniqueness because the residue and the disjoint interval component are canonical. The right formula follows by the displayed exchange of rows.

There is one algebraic child for each divisor of (d, e) , proving the multiplicity assertion. Reversing the two divisor sums in the definition of $\lambda_{G_X^{\text{row}}}$ proves (29). On physical support, d and u are squarefree and target primitivity gives $(d, u) = 1$. Thus

$$\mu(v) \mu(d/v) = \mu(d), \quad \mathbf{1}_{(v, (d/v)u)=1} = 1,$$

and (30) follows from (29). Finally, the polarization labels and disjoint integer interval convention prove (iv) and the claimed child parametrization. $\square$

Corollary 3.4 (Constructed ledger is lossless). *The native ledger Θ_X constructed above is lossless in the following concrete sense:*

- (i) *it contains the left and right polarizations once each;*
- (ii) *every row-gcd child retains its coefficient, its projector divisor, and the inverse source atom (27) or (28);*
- (iii) *the maps $\theta, t \mapsto (M_\theta(t), n_\theta, j_\theta)$ and $\theta, t \mapsto G_\theta(t)$ are recorded;*
- (iv) *all masks and weights are retained in $c_{\theta,X} \chi_\theta(t) W_{\theta,X}(t)$; and*
- (v) *summing the ledger gives the original global coefficient, not a collection of separately normalized columns.*

Proof. Items (i)–(iii) are [theorem 3.3](#). The TPC-33 fixed-period and literal weights are evaluated on the reconstructed source in $\chi_\theta W_{\theta,X}$, which proves (iv). Equation (30), followed by the exact two-term polarization, proves (v). $\square$

3.4 The native affine formula

Theorem 3.5 (Complete decorated affine export). *Let Θ_X be the native child ledger constructed in [theorem 3.3](#), with the literal weights specified in [corollary 3.4](#). Then*

$$\boxed{\begin{aligned} \mathcal{L}_{K,R}(A_X) &= \sum_{\theta \in \Theta_X} c_{\theta,X} \sum_{t \in I_\theta} \mu(D_\theta(t)) \mu(U_\theta(t)) \chi_\theta(t) W_{\theta,X}(t) \\ &\quad \times \mathbf{1}_{G_\theta(t) \leq C_X} \mathcal{K}_{K,R,X} \left(\varepsilon_\theta \frac{M_\theta(t) - n_\theta}{G_\theta(t)} \right). \end{aligned}} \quad (31)$$

For a right-ultra key, the displayed definitions are read with the two row labels exchanged. In every key the two affine signs have determinant h_0 .

Proof. Apply [lemma 3.1](#) to each term of the exact polarization (3). Take as decoration

$$F(\alpha, \gamma, j) = \mathbf{1}_{G_{\alpha,\gamma}(j) \leq C_X} \mathcal{K}_{K,R,X} \left(\Delta_{\alpha,\gamma}^\#(j) \right).$$

The explicit source–child construction and sign reassembly in [theorem 3.3](#) give (13)–(23) with the original projector multiplicity. The retained provenance map evaluates F as

$$\mathbf{1}_{G_\theta(t) \leq C_X} \mathcal{K}_{K,R,X} \left(\varepsilon_\theta \frac{M_\theta(t) - n_\theta}{G_\theta(t)} \right).$$

By [corollary 3.4](#), summing this constructed ledger reassembles exactly the two terms of (3), and hence produces (11). The determinant identity is (14); the right polarization is symmetric. $\square$

Corollary 3.6 (Fixed- h_0 representation with an unresolved content mask). *Equation (31) is a complete fixed- h_0 affine representation. Its price is that $G_\theta(t)$, and hence the exact content and determinant phase, remains a native arithmetic function of t .*

Proof. Every term is displayed in (31), and (14) is independent of t . $\square$

Remark 3.7 (No low-rank matrix conclusion). The $X^{o(1)}$ row-gcd-projector mass is columnwise after the opposite row is fixed. [Theorem 3.5](#) does not give a low-cost projective decomposition of the full joint row-pair mask or authorize absolute summation over all columns.

4 Exact-content expansion and progression resolution

4.1 Content inversion inside the low window

The content in (20) is

$$G_\theta(t) = \gcd(\sigma_\theta U_\theta(t), P_\theta).$$

For $c \geq 1$, exact common-divisor inversion gives

$$\mathbf{1}_{G_\theta(t)=c} = \sum_{\kappa \geq 1} \mu(\kappa) \mathbf{1}_{c\kappa | \sigma_\theta U_\theta(t)} \mathbf{1}_{c\kappa | P_\theta}. \quad (32)$$

The sum is finite.

Lemma 4.1 (Phase-compatible content inversion). *For every native key θ , affine parameter t , and frequency r ,*

$$\begin{aligned} & \mathbf{1}_{G_\theta(t) \leq C_X} e_{q_X} \left(-r\varepsilon_\theta \frac{M_\theta(t) - n_\theta}{G_\theta(t)} \right) \\ &= \sum_{1 \leq c \leq C_X} \sum_{\kappa \geq 1} \mu(\kappa) \mathbf{1}_{c\kappa | P_\theta} \mathbf{1}_{c\kappa | \sigma_\theta U_\theta(t)} e_{cq_X}(-r\varepsilon_\theta(M_\theta(t) - n_\theta)). \end{aligned} \quad (33)$$

Proof. Fix c . If $c\kappa$ divides both targets, primitive target support implies $c \mid M_\theta(t) - n_\theta$, so the lifted character on the right is well defined. Summing $\mu(\kappa)$ over divisors of $G_\theta(t)/c$ vanishes unless $G_\theta(t) = c$. On the surviving term,

$$e_{cq_X}(-r\varepsilon_\theta(M_\theta(t) - n_\theta)) = e_{q_X} \left(-r\varepsilon_\theta \frac{M_\theta(t) - n_\theta}{c} \right).$$

Summing $c \leq C_X$ proves the identity. $\square$

Theorem 4.2 (Complete content-expanded low window). *With the notation of theorem 3.5,*

$$\begin{aligned} \mathcal{L}_{K,R}(A_X) &= \sum_{\theta \in \Theta_X} \sum_{1 \leq c \leq C_X} \sum_{\kappa \geq 1} \sum_{r \in \mathcal{W}_X(R)} c_{\theta,X} \mu(\kappa) \mathfrak{m}_{K,X}(r) \mathbf{1}_{c\kappa | P_\theta} \\ &\quad \times \sum_{t \in I_\theta} \mu(D_\theta(t)) \mu(U_\theta(t)) \chi_\theta(t) W_{\theta,X}(t) \mathbf{1}_{c\kappa | \sigma_\theta U_\theta(t)} \\ &\quad \times e_{cq_X}(-r\varepsilon_\theta(M_\theta(t) - n_\theta)). \end{aligned} \quad (34)$$

Every sum is finite. In particular, the right side is one coherent outer sum; the (c, κ, r) terms have not been replaced by their absolute values.

Proof. Expand $\mathcal{K}_{K,R,X}$ in (31), apply lemma 4.1, and interchange finite sums. $\square$

Remark 4.3 (Why c and κ are different keys). The integer c is the candidate exact content and determines the phase modulus cq_X . The integer κ is a signed inclusion–exclusion variable. Replacing c by $b = c\kappa$ in the phase would change the normalized determinant character.

4.2 Solving the native divisibility condition

Fix an active tuple (θ, c, κ) , put

$$b = c\kappa, \quad g_{\theta,b} = (b, \sigma_\theta), \quad B_{\theta,b} = \frac{b}{g_{\theta,b}}. \quad (35)$$

Lemma 4.4 (One progression or none). *The native divisibility condition satisfies*

$$b \mid \sigma_\theta U_\theta(t) \iff B_{\theta,b} \mid U_\theta(t). \quad (36)$$

Moreover:

(i) if $(a_\theta, B_{\theta,b}) > 1$, there is no solution;

(ii) if $(a_\theta, B_{\theta,b}) = 1$, there is one residue $\tau_{\theta,b} \pmod{B_{\theta,b}}$ such that every solution is

$$t = \tau_{\theta,b} + B_{\theta,b}z, \quad z \in \mathbb{Z}.$$

Proof. Write $b = gB$ and $\sigma_\theta = g\sigma'$. By the definition of g , $(B, \sigma') = 1$. Therefore

$$gB \mid g\sigma'U_\theta(t) \iff B \mid U_\theta(t),$$

which proves (36).

The inherited primitive relations imply $(a_\theta, u_\theta) = 1$: a common divisor would divide $\sigma_\theta u_\theta - a_\theta d_\theta = h_0$, contrary to $(a_\theta, h_0) = 1$. Hence if a prime divides both a_θ and B , it does not divide u_θ , and the congruence $u_\theta + a_\theta t \equiv 0 \pmod{B}$ is insoluble. When $(a_\theta, B) = 1$, the slope is invertible modulo B , giving one residue class. $\square$

On the soluble branch, define

$$I_{\theta,b}^* = \{z \in \mathbb{Z} : \tau_{\theta,b} + B_{\theta,b}z \in I_\theta\}, \quad (37)$$

$$D_{\theta,b}^*(z) = D_\theta(\tau_{\theta,b} + B_{\theta,b}z), \quad (38)$$

$$U_{\theta,b}^*(z) = U_\theta(\tau_{\theta,b} + B_{\theta,b}z), \quad (39)$$

$$M_{\theta,b}^*(z) = \ell_\theta v_\theta D_{\theta,b}^*(z). \quad (40)$$

The set $I_{\theta,b}^*$ is an interval for each interval component already contained in θ , possibly empty. Put

$$\Omega_{\theta,b} = \ell_\theta v_\theta \sigma_\theta B_{\theta,b}. \quad (41)$$

Proposition 4.5 (Resolved phase formula). *On a soluble branch,*

$$\begin{aligned} e_{c_{qX}}(-r\varepsilon_\theta(M_{\theta,b}^*(z) - n_\theta)) \\ = \zeta_{\theta,b,c,r} e_{c_{qX}}(-r\varepsilon_\theta\Omega_{\theta,b}z), \end{aligned} \quad (42)$$

where

$$\zeta_{\theta,b,c,r} = e_{c_{qX}}(-r\varepsilon_\theta(M_\theta(\tau_{\theta,b}) - n_\theta))$$

has modulus one.

Proof. The moving row satisfies

$$M_{\theta,b}^*(z) = M_\theta(\tau_{\theta,b}) + \ell_\theta v_\theta \sigma_\theta B_{\theta,b}z.$$

Insert this identity in the character and separate the constant phase. $\square$

Corollary 4.6 (Fully resolved window). *The right side of (34) may be restricted to tuples satisfying*

$$b = c\kappa \mid P_\theta, \quad (a_\theta, B_{\theta,b}) = 1,$$

and the inner t -sum replaced exactly by

$$\begin{aligned} \zeta_{\theta,b,c,r} \sum_{z \in I_{\theta,b}^*} \mu(D_{\theta,b}^*(z)) \mu(U_{\theta,b}^*(z)) \chi_\theta(\tau_{\theta,b} + B_{\theta,b}z) \\ \times W_{\theta,X}(\tau_{\theta,b} + B_{\theta,b}z) e_{c_{qX}}(-r\varepsilon_\theta\Omega_{\theta,b}z). \end{aligned} \quad (43)$$

No content or frequency term has been omitted.

Proof. Use lemma 4.4 and proposition 4.5 in (34). Insoluble and empty branches contribute zero. $\square$

5 Content-factor extraction and fixed-determinant preservation

The resolved formula (43) still uses the value $U_{\theta,b}^*(z)$. The defining congruence forces a factor $B_{\theta,b}$ from that value. Extracting it is essential before invoking the fixed-determinant local theory of TPC-90.

5.1 The two-stage determinant law

On a soluble branch, put $B = B_{\theta,b}$, $\tau = \tau_{\theta,b}$, and define

$$V_{\theta,b}(z) = \frac{U_{\theta}(\tau + Bz)}{B}. \quad (44)$$

This is integral because $B \mid U_{\theta}(\tau)$. More explicitly,

$$\mathcal{D}_{\theta,b}(z) := D_{\theta}(\tau + Bz) = D_{\theta}(\tau) + \sigma_{\theta}Bz, \quad (45)$$

$$V_{\theta,b}(z) = \frac{U_{\theta}(\tau)}{B} + a_{\theta}z. \quad (46)$$

Theorem 5.1 (Content-progression determinant law). *Let a nonempty literal resolved term satisfy*

$$b = c\kappa \mid P_{\theta}, \quad b \mid \sigma_{\theta}U_{\theta}(t), \quad B = \frac{b}{(b, \sigma_{\theta})}.$$

Then:

(i) *the unreduced reparametrized pair*

$$\mathcal{D}_{\theta,b}(z), \quad U_{\theta}(\tau + Bz)$$

has coefficient determinant Bh_0 ;

(ii) *the second value factors exactly as*

$$U_{\theta}(\tau + Bz) = B V_{\theta,b}(z);$$

(iii) *the reduced pair*

$$\mathcal{D}_{\theta,b}(z) = D_{\theta}(\tau) + (\sigma_{\theta}B)z, \quad V_{\theta,b}(z) = \frac{U_{\theta}(\tau)}{B} + a_{\theta}z$$

has coefficient determinant exactly h_0 :

$$\boxed{(\sigma_{\theta}B) \frac{U_{\theta}(\tau)}{B} - a_{\theta}D_{\theta}(\tau) = h_0;} \quad (47)$$

(iv) *on every nonempty literal branch,*

$$(a_{\theta}, \sigma_{\theta}B) = 1, \quad (a_{\theta}\sigma_{\theta}B, h_0) = 1; \quad (48)$$

(v) *the reduced forms are individually primitive:*

$$\gcd(D_{\theta}(\tau), \sigma_{\theta}B) = 1, \quad \gcd(U_{\theta}(\tau)/B, a_{\theta}) = 1. \quad (49)$$

Proof. The unreduced forms have origins $D_{\theta}(\tau), U_{\theta}(\tau)$ and slopes $\sigma_{\theta}B, a_{\theta}B$. Their determinant is

$$(\sigma_{\theta}B)U_{\theta}(\tau) - (a_{\theta}B)D_{\theta}(\tau) = B\{\sigma_{\theta}U_{\theta}(\tau) - a_{\theta}D_{\theta}(\tau)\} = Bh_0.$$

The progression was chosen so that $B \mid U_{\theta}(\tau)$, proving the factorization. Dividing the second origin and slope by B gives (47).

For a nonempty term, b divides both physical targets. Primitive target support therefore gives

$$(b, h_0) = 1 \quad (50)$$

and, on every contributing source atom, b is coprime to the moving row and j_{θ} . In particular $(B, h_0) = 1$. Solubility gives $(a_{\theta}, B) = 1$ by lemma 4.4; combine this with $(a_{\theta}, \sigma_{\theta}) = 1$ and $(a_{\theta}\sigma_{\theta}, h_0) = 1$ from (14) to obtain (48).

Any common divisor of $D_{\theta}(\tau)$ and $\sigma_{\theta}B$ divides the left side of (47), hence h_0 ; this contradicts $(\sigma_{\theta}B, h_0) = 1$. The second primitive identity is proved in the same way using $(a_{\theta}, h_0) = 1$. $\square$

5.2 Exact Möbius extraction

Lemma 5.2 (Forced content factor in the Möbius sign). *For $B \geq 1$ and $V \geq 1$,*

$$\mu(BV) = \begin{cases} \mu(B)\mu(V)\mathbf{1}_{(B,V)=1}, & \mu^2(B) = 1, \\ 0, & \mu^2(B) = 0. \end{cases} \quad (51)$$

Proof. If B is not squarefree, then BV is not squarefree. If B is squarefree and $(B, V) > 1$, again BV has a squared prime factor. On the remaining branch, multiplicativity of μ gives the identity. $\square$

Theorem 5.3 (Fixed- h_0 reduced native export). *In the fully resolved low-window formula, every branch with $\mu^2(B_{\theta,b}) = 0$ vanishes. On the remaining branches, (43) equals*

$$\begin{aligned} \mu(B_{\theta,b})\zeta_{\theta,b,c,r} \sum_{z \in I_{\theta,b}^*} \mu(\mathcal{D}_{\theta,b}(z))\mu(V_{\theta,b}(z))\mathbf{1}_{(B_{\theta,b}, V_{\theta,b}(z))=1} \\ \times \chi_{\theta}(\tau_{\theta,b} + B_{\theta,b}z)W_{\theta,X}(\tau_{\theta,b} + B_{\theta,b}z) \\ \times e_{cq_X}(-r\varepsilon_{\theta}\Omega_{\theta,b}z). \end{aligned} \quad (52)$$

The two Möbius arguments in (52) are individually primitive affine forms with the same prescribed determinant h_0 . All extracted factors and masks remain literal.

Proof. Use

$$U_{\theta,b}^*(z) = B_{\theta,b}V_{\theta,b}(z)$$

in (43) and apply lemma 5.2. The determinant and primitive claims are theorem 5.1. $\square$

Corollary 5.4 (Exact TPC-90 local split after content extraction). *For every reduced branch in theorem 5.3, the TPC-90 identity applies:*

$$\begin{aligned} \mu(\mathcal{D}_{\theta,b}(z))\mu(V_{\theta,b}(z)) \\ = \mu^2(\mathcal{D}_{\theta,b}(z))\mu^2(V_{\theta,b}(z))\lambda(\mathcal{D}_{\theta,b}(z))\lambda(V_{\theta,b}(z)). \end{aligned} \quad (53)$$

Its simultaneous squarefree local root counts are those for a primitive determinant- h_0 pair. The extra mask $\mathbf{1}_{(B_{\theta,b}, V_{\theta,b}(z))=1}$, the sign $\mu(B_{\theta,b})$, and the additive determinant character remain outside that identity.

Proof. Apply the exact identity $\mu = \mu^2\lambda$ to the two positive reduced forms. The hypotheses of the TPC-90 primitive local classification follow from theorem 5.1. $\square$

5.3 The precise provenance boundary

We now isolate the exact obstruction to an untracked determinant change.

Theorem 5.5 (No silent determinant relabeling). *After the substitution $t = \tau + Bz$, the unreduced pair has determinant Bh_0 . Therefore it cannot be entered into a theorem whose hypothesis is determinant h_0 by simply retaining its two unreduced values and changing the determinant label. A valid fixed- h_0 entry must instead perform*

$$U(\tau + Bz) = BV(z)$$

and retain:

- (a) the branch deletion $\mu^2(B) = 0$;
- (b) the outer sign $\mu(B)$;
- (c) the coprimality mask $\mathbf{1}_{(B,V(z))=1}$; and
- (d) the original content, phase, polarization, and outer provenance keys.

Dropping any one of (a)–(c) changes the literal coefficient.

Proof. The determinant statements follow from [theorem 5.1](#). The three coefficient effects are the exhaustive cases of (51). The remaining provenance is needed to reverse the finite expansion in [theorem 4.2](#); without it one no longer has the source coefficient (5). $\square$

Example 5.6 (The two determinants are visibly different). Take

$$D(t) = t, \quad U(t) = t + 1, \quad h_0 = 1.$$

Restrict to $2 \mid U(t)$, so $t = 1 + 2z$. The unreduced pair

$$1 + 2z, \quad 2 + 2z$$

has determinant 2. Extracting the forced factor from the second value gives

$$1 + 2z, \quad 1 + z,$$

whose determinant is 1. The equality

$$\mu(2 + 2z) = -\mu(1 + z)\mathbf{1}_{(2,1+z)=1}$$

shows why the sign and coprimality mask cannot be omitted. This is a finite algebraic example, not a model of the complete physical packet.

6 An exact short/resonant/generic partition

6.1 Resolved native indices

Let $\Xi_X(K, R)$ be the set of all tuples

$$\xi = (\theta, c, \kappa, r) \tag{54}$$

that occur in [theorem 5.3](#), including:

$$\begin{aligned} 1 \leq c \leq C_X, \quad r \in \mathcal{W}_X(R), \quad b_\xi = c\kappa \mid P_\theta, \\ \mu^2(B_\xi) = 1, \quad (a_\theta, B_\xi) = 1, \quad I_\xi := I_{\theta, b_\xi}^* \neq \emptyset. \end{aligned}$$

Write

$$N_\xi = \#I_\xi, \tag{55}$$

$$\Omega_\xi = \ell_\theta v_\theta \sigma_\theta B_\xi, \tag{56}$$

$$\delta_\xi = \left\| \frac{r\Omega_\xi}{cq_X} \right\|_{\mathbb{R}/\mathbb{Z}}. \tag{57}$$

The complete outer coefficient is

$$\mathfrak{c}_{\xi, X} = c_{\theta, X} \mu(\kappa) \mu(B_\xi) \mathfrak{m}_{K, X}(r) \zeta_{\theta, b_\xi, c, r}. \tag{58}$$

Put

$$\begin{aligned} F_{\xi, X}(z) &= \mu(\mathcal{D}_{\theta, b_\xi}(z)) \mu(V_{\theta, b_\xi}(z)) \mathbf{1}_{(B_\xi, V_{\theta, b_\xi}(z))=1} \\ &\quad \times \chi_\theta(\tau_{\theta, b_\xi} + B_\xi z) W_{\theta, X}(\tau_{\theta, b_\xi} + B_\xi z). \end{aligned} \tag{59}$$

Then the fully resolved exact formula is

$$\mathcal{L}_{K, R}(A_X) = \sum_{\xi \in \Xi_X(K, R)} \mathfrak{c}_{\xi, X} \sum_{z \in I_\xi} F_{\xi, X}(z) e_{cq_X}(-r\varepsilon_\theta \Omega_\xi z). \tag{60}$$

All variables on the right are components of ξ ; the repeated symbols c, r, θ are not newly summed inside the inner sum.

Proposition 6.1 (Resolved occupancy). *Every resolved interval satisfies*

$$N_\xi \leq 1 + \frac{|I_\theta|}{B_\xi} \ll 1 + \frac{D_X^{\text{row}}}{\sigma_\theta v_\theta B_\xi}. \tag{61}$$

Proof. The values $t = \tau_{\theta, b} + B_\xi z$ are spaced by B_ξ inside I_θ . Apply (15). $\square$

6.2 The three sectors

Let $H_X \geq 1$ be any chosen occupancy threshold. Define disjoint sets:

$$\Xi_{\text{short}} = \{\xi : N_\xi \leq H_X\}, \quad (62)$$

$$\Xi_{\text{res}} = \{\xi : N_\xi > H_X, N_\xi \delta_\xi \leq 1\}, \quad (63)$$

$$\Xi_{\text{gen}} = \{\xi : N_\xi > H_X, N_\xi \delta_\xi > 1\}. \quad (64)$$

Call them the short, long resonant, and long generic sectors. The resonant sector includes both exact resonances and phases that move by at most one turn at the natural fiber resolution.

Theorem 6.2 (Complete native sector decomposition). *For every $H_X \geq 1$,*

$$\Xi_X(K, R) = \Xi_{\text{short}} \dot{\cup} \Xi_{\text{res}} \dot{\cup} \Xi_{\text{gen}}, \quad (65)$$

and

$$\boxed{\mathcal{L}_{K,R}(A_X) = \mathcal{L}_{\text{short},X} + \mathcal{L}_{\text{res},X} + \mathcal{L}_{\text{gen},X}}, \quad (66)$$

where each term is (60) restricted to the corresponding set. This identity retains both polarizations, every exact-content projector term, every low frequency, and every resolved fiber.

Proof. Every ξ first satisfies exactly one of $N_\xi \leq H_X$ and $N_\xi > H_X$. In the latter case it satisfies exactly one of $N_\xi \delta_\xi \leq 1$ and $N_\xi \delta_\xi > 1$. Restrict the finite sum (60) to the resulting disjoint sets. $\square$

Proposition 6.3 (One-step and observed resonance criteria). *The one-step multiplier of the determinant character equals one precisely when*

$$\delta_\xi = 0 \iff cq_X \mid r\ell_\theta v_\theta \sigma_\theta B_\xi. \quad (67)$$

If $N_\xi \geq 2$, this is also equivalent to the character being constant on the resolved interval I_ξ . If $N_\xi = 1$, the character is vacuously constant on its single observed point for every value of δ_ξ ; such a singleton belongs to the short sector for every $H_X \geq 1$.

For all sufficiently large X , $c < q_X$. Since $0 < d_{q_X}(r, 0) < q_X/2$, an exact resonance in particular requires

$$q_X \mid \ell_\theta v_\theta \sigma_\theta B_\xi.$$

No theorem presently excludes that divisibility on every literal key.

Proof. The phase increment from z to $z + 1$ is

$$e_{cq_X}(-r\varepsilon_\theta \Omega_\xi).$$

It equals one exactly under the divisibility in (67). An integer interval with at least two points contains two consecutive integers, which proves the observed constancy assertion. The singleton assertion is immediate from (62). The critical scales give $c \leq C_X < q_X$, and the chosen representative of a nonzero low frequency is not divisible by the prime q_X . Euclid's lemma gives the final necessary condition. $\square$

Remark 6.4 (Near resonance is not exact resonance). The condition $N_\xi \delta_\xi \leq 1$ is quantitative and depends on the actual occupancy. It does not assert the exact divisibility in (67). It records that additive oscillation alone is too slow on that particular interval to be treated as a uniformly generic phase.

6.3 Deterministic sector budgets

For $\mathcal{A} \in \{\text{short}, \text{res}, \text{gen}\}$, define the literal absolute envelope

$$\mathfrak{B}_{\mathcal{A},X} = \sum_{\xi \in \Xi_{\mathcal{A}}} |\mathfrak{c}_{\xi,X}| \sum_{z \in I_{\xi}} |F_{\xi,X}(z)|. \quad (68)$$

Proposition 6.5 (Safe but possibly expensive return). *For every sector,*

$$|\mathcal{L}_{\mathcal{A},X}| \leq \mathfrak{B}_{\mathcal{A},X}. \quad (69)$$

Consequently,

$$|\mathcal{L}_{K,R}(A_X)| \leq \mathfrak{B}_{\text{short},X} + \mathfrak{B}_{\text{res},X} + \mathfrak{B}_{\text{gen},X}. \quad (70)$$

These inequalities are sufficient return estimates, not claims that the envelopes have the required physical size.

Proof. Apply the triangle inequality after the exact decomposition (66); all characters have modulus one. $\square$

Remark 6.6 (Where coherence may be lost). Although $\sum_r |\mathfrak{m}_{K,X}(r)| \ll_K 1$, absolute summation over (c, κ) , native columns, or polarizations can destroy projector and outer-row cancellation. The bounded Wiener norm therefore does not imply that (70) is subpower relative to the physical target.

Remark 6.7 (Generic phase is not a signed estimate). The inequality $N_{\xi} \delta_{\xi} > 1$ says that the additive character makes more than one resolvable turn. The coefficient $F_{\xi,X}$ still contains two Möbius/Liouville signs, masks, and varying weights. No geometric-series bound applies unless a future theorem controls the partial sums or variation of that coefficient.

7 The literal arithmetic gate and endpoint ledger

7.1 What the export reduces

At the critical scales, choose

$$R_X = X^{\varrho+o(1)} < q_X/2, \quad \varrho < \frac{267}{400},$$

and fixed $K \geq 2$. For a desired zero-mode rate

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2,$$

the inherited high-frequency tail has a fixed-power reserve when

$$(K - \tfrac{1}{2})\varrho > \frac{133}{400} + \eta_Z. \quad (71)$$

The often quoted second-order choice

$$K = 2, \quad R_X = X^{133/600+\varepsilon}$$

belongs to the zero-loss corner $\eta_Z = 0$ for fixed

$$0 < \varepsilon < \frac{107}{240}.$$

For the same window and positive η_Z , one additionally needs $\eta_Z < 3\varepsilon/2$.

Theorem 7.1 (Exact sufficient sector certificate). *Assume (71). If a future theorem proves*

$$|\mathcal{L}_{\text{short},X} + \mathcal{L}_{\text{res},X} + \mathcal{L}_{\text{gen},X}| \leq X^{-\eta_Z+o(1)} Q_X^2, \quad (72)$$

then

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2.$$

A stronger sufficient condition is

$$|\mathcal{L}_{\text{short},X}| + |\mathcal{L}_{\text{res},X}| + |\mathcal{L}_{\text{gen},X}| \leq X^{-\eta_Z+o(1)} Q_X^2. \quad (73)$$

Neither estimate is proved here.

Proof. Use (66) in (12). Under (71), the tail is smaller than the target by a fixed power. The second displayed hypothesis implies the first by the triangle inequality. $\square$

Remark 7.2 (The coherent certificate is weaker). Equation (72) permits cancellation between the three returned sectors. Requiring three separate bounds may lose a fixed power. A proof may use either route, but must state which outer cancellations survive.

7.2 Sector-specific tasks

Short fibers. By [proposition 6.1](#), large values of $\sigma_\theta v_\theta B_\xi$ automatically enter the short sector. These fibers have no asymptotic averaging parameter. They must be returned by an exact enumeration, an outer-energy estimate, or a physical counting theorem. Calling them negligible because each is short does not bound their number or outer coefficient mass.

Long resonant fibers. Exact resonances satisfy (67); near resonances satisfy $N_\xi \delta_\xi \leq 1$. The additive determinant phase provides no uniform power saving on this sector. A viable estimate must instead use the two-sign arithmetic, a structural outer cancellation, or a proved small multiplier/provenance mass.

Long generic fibers. Here the additive phase makes more than one resolvable turn, and the reduced Möbius pair has the fixed determinant h_0 . This is the natural sector for a large-sieve, bilinear, TT^* , or phase-sensitive affine-correlation theorem. Such a theorem still must be uniform in the growing slopes, the content-factor masks, prefixes, interval origins, and complete outer ledger.

Local squarefree split. [Corollary 5.4](#) applies TPC-90 without changing the determinant. It classifies local square-power support but leaves the weighted Liouville product. The extracted $\mu(B_\xi)$ may participate in outer cancellation, but no such cancellation is proved by the factorization itself.

7.3 A provenance success certificate

The following definition records the minimum data that a later arithmetic estimate must return.

Definition 7.3 (Callable low-window theorem). A future estimate is callable for the literal TPC-86 gate only if it returns:

- (i) the same filter multiplier $\mathbf{m}_{K,X}(r)$ and every $r \in \mathscr{W}_X(R_X)$;
- (ii) both polarizations and the complete constructed source–child ledger Θ_X , including its inverse provenance and projector multiplicity;
- (iii) every $c, \kappa, B, \mu(B)$, content mask, and exact phase;
- (iv) the short, resonant, and generic sectors, either separately or through their complete coherent sum;

- (v) the same fixed nonzero h_0 , interval origins, prefixes, weights, and one global post-aggregation normalization; and
- (vi) a quantitative outer rate compatible with (71) and the physical endpoint ledger.

Proposition 7.4 (Downgrade conditions). *An estimate is not the literal low-window certificate if it:*

- (a) *treats only one generic frequency or one fixed affine column;*
- (b) *deletes exact or near resonances;*
- (c) *assumes every fiber length tends to infinity;*
- (d) *uses the unreduced determinant- Bh_0 pair but labels it determinant h_0 ;*
- (e) *extracts B but omits $\mu(B)$ or $\mathbf{1}_{(B,V)=1}$;*
- (f) *sums an exceptional-set cardinality without its reconstruction multiplier and outer coefficients; or*
- (g) *leaves an unreturned average over shifts, scales, origins, or native keys.*

Such a result may remain valid at a restricted, frozen, averaged, or negative-transfer level.

Proof. Each item changes or omits a term in the exact master formula (60) or in the fixed- h_0 reduction (52). It therefore estimates a different coefficient unless a separate return theorem is supplied. $\square$

7.4 Independent ledgers

If future work proves

$$D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2$$

and the zero-mode bound at rate η_Z , the determinant compatibility remains

$$\boxed{\lambda_D \leq 2\eta_Z.} \tag{74}$$

Separately, every fixed-power loss introduced by the sector partition, coefficient amplification, short-fiber return, masks, outer reassembly, and downstream physical comparison, excluding losses already recorded in $\lambda_D - 2\eta_Z$, must satisfy

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}.} \tag{75}$$

Equality in (74) has no determinant reserve; equality in (75) stops the physical route. The same cost is not entered in both ledgers.

8 Claim boundary and conclusion

Theorem 8.1 (Audited Deliverable 1A conclusion). *For the literal high-row TPC-32 coefficient at one prescribed nonzero h_0 , the complete TPC-86 multiplier-weighted low-frequency window admits:*

- (i) *the exact row-native formula (11);*
- (ii) *the constructed, lossless two-polarization affine formula (31);*
- (iii) *the exact-content expansion (34);*
- (iv) *the one-progression resolution (43);*
- (v) *the exact content-factor extraction (52), whose reduced affine pair has determinant h_0 ; and*
- (vi) *the disjoint short/resonant/generic decomposition (66).*

No native key, low frequency, exact-content term, resonance, bounded fiber, or polarization is deleted in these identities.

Proof. Items (i)–(vi) are theorems 2.1, 3.5, 4.2, 5.3 and 6.2 and corollary 4.6, respectively. $\square$

What is new. The paper proves three exact advances. First, the full coherent TPC-86 window is now written in the native affine variables rather than represented by a schematic twisted column. Second, exact content resolution is reconciled with the fixed determinant: the unreduced progression has determinant Bh_0 , but the forced factor B in one Möbius argument can be extracted exactly, restoring determinant h_0 together with the literal sign and coprimality mask. Third, the resulting physical family is partitioned into explicit short, resonant, and generic sectors with no exceptional remainder.

Proof levels. The Fourier reindexing, phase-compatible content inversion, progression solution, Möbius extraction, determinant identities, and sector partition are finite L0 theorems. Their attachment to the already aggregated TPC-32 coefficient through the actual TPC-33 column decomposition is an L1 theorem.

Finite regression certificate. The standard-library script `experiments/verify_literal_export.py` separately exercises both source polarizations, the explicit source-child inverse, projector and sign reassembly, phase-compatible content inversion, progression resolution, determinant reduction, and Möbius extraction on deterministic finite instances. This guards the literal bookkeeping; it is not evidence for an L2 estimate.

What is not proved. This paper proves no nontrivial upper bound for any of

$$\mathcal{L}_{\text{short},X}, \quad \mathcal{L}_{\text{res},X}, \quad \mathcal{L}_{\text{gen},X},$$

no new bound for Z_X , no post-bin mass or determinant-energy lower bound, and no growing fixed- h_0 Liouville/Möbius correlation estimate. It proves no L2 saving.

The surviving arithmetic fork. The next positive step may:

- (a) control short fibers by their complete outer mass or an exact physical return;
- (b) bound the multiplier/provenance mass of resonant fibers or use the fixed- h_0 sign correlation there;
- (c) prove a phase-sensitive estimate for the long generic reduced pair; or
- (d) estimate the complete signed sum of all three sectors without separate absolute bounds.

Any one-sector theorem is progress at its native level only after the other two sectors are returned.

Stop rules. A generic-frequency theorem does not pass if it leaves resonant multiplier mass uncontrolled. A long-fiber theorem does not pass if the aggregate bounded-fiber contribution is unreturned. A determinant- h_0 theorem cannot be applied to the unreduced determinant- Bh_0 pair. After extracting B , the factors $\mu(B)$ and $\mathbf{1}_{(B,V)=1}$ are mandatory. Any unavoidable fixed-power return loss reaching $1/400$ stops the physical branch.

Independent requirements. The zero-mode and determinant-energy exponents still require

$$\lambda_D \leq 2\eta Z.$$

The separate physical ledger still requires

$$\Lambda_{\text{phys}} < 1/400.$$

TPC-18 full-block reassembly and primitive determinant dispersion remain independent downstream gates; the affine export proves neither one.

Final boundary. Nothing is specialized to $h_0 = 2$. The paper does not prove a fixed-shift Hardy–Littlewood asymptotic, breach the sieve parity barrier, give a prime-pair lower bound, or prove the twin-prime conjecture. Its contribution is a complete literal representation and a corrected fixed-determinant content reduction on which the next arithmetic estimate can be stated without hidden sectors.

References

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